

Can a High School Personal Finance Course Reduce the Gender Gap in Being Unbanked?

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Abstract

Among unpartnered young adults in the United States—whose banking status reflects their own financial decisions rather than a partner’s—women are significantly more likely than men to be unbanked, even after accounting for income and education. I determine whether state-mandated high school personal finance courses reduce this gender gap, and if so, through what mechanisms. Using the FDIC National Survey of Unbanked and Underbanked Households (2009–2023) and a heterogeneity-robust difference-in-differences estimator, I estimate that mandates reduce the likelihood of being unbanked for both women and men, but by significantly more for women, narrowing the gender gap by 41 percent. The effects are concentrated among women with no education beyond high school and among Black and Hispanic women—groups that face the greatest barriers to financial inclusion. I examine two mechanisms, and the evidence supports both. First, the mandate increases women’s objective financial knowledge, although not their self-assessed knowledge. Second, it improves economic outcomes by increasing college completion, self-employment, and the likelihood of attaining middle-income status. Because it reaches individuals before financial independence and does not rely on voluntary participation, mandatory high school financial education can reduce persistent gender disparities in financial inclusion that adult financial education programs have struggled to address.

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1 Introduction

Bank accounts are important gateways to participation in the formal financial system. They allow households to receive income, make payments, accumulate savings, establish credit histories, and access mainstream financial products. However, despite the widespread availability of banking services in the United States, approximately 4.22 percent of households, or about 5.6 million households, remained unbanked in 2023.¹ Among young adults without partners, the unbanked share rises to 10.33 percent. Moreover, within this subgroup, where banking decisions largely reflect individuals' own financial choices rather than those of a partner, a staggering 13.28 percent of women are unbanked, compared to 7.12 percent of men. Much of this gap reflects the fact that single women tend to be worse off financially than their male counterparts. However, even after accounting for differences in income, education, and other observable characteristics, a substantial gap remains, suggesting that factors beyond economic resources continue to limit women's access to the banking system. Understanding the sources of this persistent disparity is important for identifying policies that can effectively promote financial inclusion.

In this paper, I ask whether public policy can reduce the gender gap in financial inclusion, by studying the effects of mandatory high school personal finance education on the likelihood of being unbanked. Existing research describes who is unbanked and why (Bogan and Wolfolds, 2022; Hayashi, Routh and Toh, 2024), but much less is known about the policies that could close these gaps. One reason to expect financial education to help is that opening and maintaining a bank account requires understanding financial products, comparing account options, and navigating financial institutions, so improving financial knowledge before adulthood can raise banking participation and narrow the gender gap.

The analysis combines a descriptive decomposition with a causal research design. I first examine how much of the gender gap in being unbanked can be explained by differences between unpartnered women and men in income, education, household structure, and other

¹Author's calculations using the 2023 FDIC National Survey of Unbanked and Underbanked Households.

observable characteristics. Using a Kitagawa–Oaxaca–Blinder decomposition of the gender gap in unbanked status using data from the Federal Deposit Insurance Corporation (FDIC) survey, I quantify the contribution of each characteristic. Because this decomposition is descriptive and cannot establish whether changing these characteristics would narrow the gap, I then turn to a specific policy: state mandates requiring students to complete a personal finance course before graduating from high school. Unlike voluntary financial education programs for adults, these mandates reach nearly all students before they become financially independent, avoiding concerns about selective participation. Variation in the timing of mandate adoption across states and graduating cohorts provides the basis for identifying the policy’s effect. I combine information on mandate timing with data from the 2009 through 2023 waves of the FDIC National Survey of Unbanked and Underbanked Households and estimate the policy’s effect using the extended two-way fixed-effects estimator of Wooldridge (2025), which accommodates heterogeneous treatment effects under staggered policy adoption.

I then examine the mechanisms through which the mandates affect banking outcomes. One possibility is that they improve financial literacy, making young adults better equipped to open and manage bank accounts. Another is that they improve economic circumstances by increasing educational attainment, employment, and income, thereby making bank account ownership more accessible and valuable. I examine the financial literacy channel using the National Financial Capability Study, which includes the Big Five financial literacy questions absent from the FDIC survey, and the economic channel by estimating the mandates’ effects on educational attainment, employment, and income using the FDIC survey.

I find that mandatory high school personal finance education narrows the gender gap in being unbanked. Exposure lowers men’s probability of being unbanked by 1.23 percentage points, or about 17 percent of the rate among men, and women’s by 3.76 percentage points, or about 28 percent of the rate among women, closing roughly 41 percent of the gender gap. The effects are concentrated among women with only a high school education and

among Black and Hispanic women, the groups furthest from the formal financial system to begin with. The evidence supports both mechanisms. Relative to men, exposure increases women’s financial knowledge scores by 0.13 points, about 6 percent of the mean, with the gains concentrated in the two investment questions. In contrast, it has no effect on women’s self-assessed financial knowledge, suggesting that what changes is what they know rather than what they believe they know. Women’s economic circumstances also improve. Relative to men, exposure makes women 2.71 percentage points more likely to complete college, 1.33 percentage points more likely to be self-employed without incorporating, and 3.85 percentage points more likely to have income between \$50,000 and \$75,000.

These findings suggest that mandatory high school personal finance education can be an effective policy tool for expanding financial inclusion among young adults by reducing barriers to banking participation. The larger gains among women indicate that broad-based financial education can also narrow gender disparities in financial inclusion. At the same time, the weaker effects for men suggest that a uniform curriculum may not benefit all students equally, highlighting the importance of accounting for heterogeneous responses when designing financial education policies.

2 Related Literature

The first body of research examines the causes and consequences of being unbanked. This literature has established that households without a bank account are disproportionately low-income, less educated, and members of racial and ethnic minority groups (Boel and Zimmerman, 2022; Calem, Henderson and Wang, 2026). Less is known about how these patterns translate into gender disparities in financial inclusion. Bogan and Wolfolds (2022) show that gender differences in being unbanked vary substantially across racial and ethnic groups: Black women are more likely to be unbanked than Black men and than any other race-gender group, Hispanic women are less likely to be unbanked than Hispanic men, and

White women and men have similar rates. Because existing measures of banking status are collected at the household level, however, these comparisons may reflect both individual financial decisions and differences in who reports household financial information. Focusing on unpartnered adults, for whom household banking status corresponds more closely to individual banking participation, I document in Section 5 that women are substantially more likely than men to be unbanked.

A growing literature examines the factors underlying these disparities and the potential for policy to reduce them. Calem, Henderson and Wang (2026) show that the decline in the overall unbanked rate has coincided with rising incomes, but that remaining unbanked households are increasingly concentrated among single individuals. They also find that states with higher levels of financial literacy have lower unbanked rates, suggesting that financial capability may be an important determinant of banking participation. However, evidence on whether policies that improve financial knowledge can expand financial inclusion remains limited. Policies that focus solely on increasing access to bank accounts have generally produced limited long-run effects (Dupas et al., 2018; Collins, Larrimore and Urban, 2026), suggesting that demand-side barriers may play an important role. Such barriers include limited financial knowledge and distrust of financial institutions (Hayashi, Routh and Toh, 2024). They are not all of one kind. Some are preference-based, in that the household does not want an account. Others are informational, in that the household wants one but does not know which products carry low or no minimum balances, or how to open and maintain one. Policies that expand access address neither, and an education mandate can address only the second. This paper contributes to this literature by examining whether a policy designed to improve financial knowledge, mandatory high school personal finance education, increases banking participation and whether its effects differ across demographic groups.

The second body of research examines financial literacy and, in particular, gender gaps in financial literacy. Across countries, women score lower than men on standard measures of financial knowledge, are more likely to respond “don’t know” rather than guess, and re-

port lower confidence in their financial knowledge even conditional on measured knowledge (Lusardi and Mitchell, 2011; Bucher-Koenen et al., 2017; Aristei and Gallo, 2022). This gap cannot be fully explained by differences in observable characteristics. Decomposition analyses show that differences in income, education, and other demographic characteristics account for only a small share of the gender gap, with much of the remaining difference reflecting variation in the relationship between these characteristics and financial knowledge (Fonseca et al., 2012; Aristei and Gallo, 2022). Consistent with this interpretation, Fonseca et al. (2012) show that financial decision-making within couples does not follow a simple specialization model in which one spouse manages household finances, but instead depends on the relative education of partners. Aristei and Gallo (2022) further suggest that behavioral traits and social norms regarding gender roles in financial decisions contribute to the remaining gap. Importantly, gender differences in financial literacy persist among groups for whom individual financial knowledge is particularly consequential, including single women and widows (Bucher-Koenen et al., 2017). These findings make financial literacy a particularly relevant demand-side barrier to study: it exhibits substantial gender disparities, is not merely a consequence of differences in income and education, and is potentially amenable to policy interventions through education. Whether improvements in financial literacy translate into greater financial inclusion, however, remains an open empirical question.

The third body of research examines the effects of state-mandated high school financial education. Early evaluations of these mandates often found limited effects on subsequent financial knowledge and behavior (Cole, Paulson and Shastry, 2016; Fernandes, Lynch and Netemeyer, 2014; Willis, 2011). More recent evidence, however, has produced more positive findings by leveraging the staggered adoption of mandates across states and cohorts. This literature shows that exposure to high school financial education increases financial knowledge and improves a range of downstream financial outcomes, including credit management, borrowing decisions, debt outcomes, and reliance on alternative financial services (Walstad, Rebeck and MacDonald, 2010; Kaiser and Menkhoff, 2020; Urban et al., 2020; Stoddard

and Urban, 2020; Brown et al., 2016; Harvey, 2019). Evidence from randomized evaluations outside the United States similarly finds that school-based financial education improves both financial knowledge and financial behaviors (Bruhn et al., 2016, 2025; Frisanco, 2023). Together, these studies demonstrate that financial education mandates can influence financial capability and a broad range of financial outcomes. However, less is known about whether such policies increase participation in the formal banking system and whether their effects differ across demographic groups.

A small but growing literature examines the relationship between financial education and banking participation directly. Harvey (2019) finds that financial education reduces young adults’ reliance on alternative financial services, partly through improvements in financial literacy and financial planning. More recently, Routh and Urban (2025) use the FDIC National Survey of Unbanked and Underbanked Households to show that state financial education mandates reduce the likelihood of being unbanked, particularly in states with stronger payday lending restrictions. Their findings highlight the potential role of financial education in expanding financial inclusion, but their analysis focuses on the interaction between financial education and payday lending regulation rather than on the direct effects of education mandates on banking participation. This paper builds on this literature by examining whether state-mandated financial education increases banking participation and whether these effects contribute to reducing gender disparities in financial inclusion.

My study addresses these gaps in three ways. First, I examine whether high school financial education reduces gender disparities in banking participation, rather than simply lowering the overall unbanked rate. To better capture individuals’ own financial decisions, I focus on unpartnered young adults, for whom household banking status more closely corresponds to individual banking participation.² Second, I examine which groups benefit most from the policy and show that the largest gains are concentrated among women with no education

²Because unbanked status is measured at the household level, married or cohabiting respondents share the same banking status as their partners. As a result, gender differences among partnered individuals may not reflect differences in individual financial decisions. I discuss this issue further in Section 4.

beyond high school and among Black and Hispanic women, groups with the lowest baseline rates of banking participation. Third, I explore why financial education has larger effects on banking participation among women than men. Using the National Financial Capability Study, I show that mandate exposure increases women’s objective financial knowledge but does not improve their self-assessed knowledge, consistent with the distinction between financial knowledge and confidence emphasized by Aristei and Gallo (2022). Using the FDIC survey, I further show that mandate exposure is associated with improvements in women’s socioeconomic outcomes through higher college completion, greater self-employment, and movement into the middle of the income distribution.³ Together, these contributions extend the literature from asking whether financial education affects financial outcomes to understanding whether it can reduce demographic disparities in financial inclusion, which groups benefit most, and the mechanisms underlying these effects.

3 Institutional Background

Over the past several decades, states have increasingly adopted legislation requiring high school students to complete personal finance coursework before graduation. By 2030, 42 states will have implemented such requirements. These mandates were designed to improve students’ financial knowledge and prepare them to manage financial decisions as they transition into adulthood. Personal finance courses are typically offered in grades 10 through 12, with most students taking them during their junior or senior year, and cover topics such as budgeting, saving, *banking*, credit, debt management, insurance, and investing. The banking component is directly relevant to the outcome studied here: it teaches students how to compare account fees, select among financial products, open accounts, and avoid costly mistakes such as overdrafts. By providing this information before students begin managing their own finances, personal finance education may influence whether young adults enter adulthood

³This finding is consistent with Struckell et al. (2022), who find a stronger positive relationship between financial literacy and self-employment among women than among men.

with a connection to the formal banking system.

These state mandates provide a useful setting for studying whether financial education affects banking participation because exposure is determined by state graduation requirements rather than individual decisions to enroll in a course. All students in an affected graduating cohort must complete the required instruction to graduate, including students who might not otherwise receive financial education at school or at home. This feature distinguishes mandatory financial education from voluntary programs, where participation may reflect unobserved differences in financial interest or ability.

A second feature of these mandates is their timing. Although some students may already have experience using a bank account before completing high school, the transition out of high school is when individuals increasingly begin managing finances independently, including earning income, paying bills, and making their own financial decisions. Personal finance courses are typically completed shortly before this transition, allowing the instruction to shape financial behaviors as young adults begin managing their finances independently.

States adopted these requirements at different times, generating variation in exposure across states and graduating cohorts. Illinois is the earliest adopter, with the requirement first applying to the 1970 graduating cohort. 3 additional states first implemented requirements for cohorts graduating in the 1990s, followed by 11 states in the 2000s and 16 states in the 2010s, with remaining states adopting in the 2020s and beyond. A summary of the timing of adoption and the first graduating cohort exposed to each requirement is provided in Table A2.

States differ in how they implement personal finance graduation requirements. Some require students to complete a stand-alone personal finance course, while others integrate personal finance instruction into existing courses such as economics, consumer education, or civics. As shown in Table A2, stand-alone requirements are uncommon over the period I study. Among the 31 states contributing treated cohorts, only 5 require a stand-alone course by the 2019 cohort, while 26 provide instruction through an embedded curriculum. Most

stand-alone requirements were adopted after the cohorts observed in my sample. I therefore define treatment as whether a state requires personal finance instruction for graduation, regardless of whether the instruction is delivered through a stand-alone course or an embedded curriculum. I measure exposure using the first graduating cohort required to complete the mandate rather than the year of legislative adoption, because implementation often begins several years after legislation is enacted.

4 Data

I use two data sources. The primary dataset is the FDIC National Survey of Unbanked and Underbanked Households, which I use to estimate the effects of state personal finance mandates, examine treatment heterogeneity, and test the economic mechanism. I supplement this analysis with the National Financial Capability Study (NFCS), which contains detailed measures of financial knowledge, including the Big Five financial literacy quiz. Although the NFCS records respondents' ages only in broad categories, making it less suitable for the main identification strategy, it is what allows me to test whether the mandates improve financial knowledge.

4.1 FDIC National Survey of Unbanked and Underbanked Households

The FDIC survey is a nationally representative supplement to the Current Population Survey (CPS), fielded every two years since 2009, and I use all eight waves available at the time of writing, 2009 through 2023. The survey is the leading source of information on banking participation in the United States, and it records household banking status together with rich demographic and socioeconomic characteristics. Because it records each respondent's state of residence and age, it can be linked to the timing of state personal finance mandates, under assumptions about where respondents attended high school that I set out below.

Sample selection criteria. The supplement is answered by one designated respondent per household, who reports banking status on the household’s behalf, so each observation is both a respondent and a household. Across the 2009 through 2023 waves, 296,005 households completed the supplement, out of 570,943 CPS households in the months it was fielded. I reduce these to the 25,404 respondents used in estimation by imposing four criteria in the following order. First, I drop 28,581 respondents who did not complete high school, because treatment is defined by exposure to a graduation requirement and those who left school before graduating were never subject to one. Second, I drop 178,613 respondents outside the 1991 through 2019 high school graduating cohorts, because cohorts graduating earlier experienced different underlying trends and make less comparable controls, and cohorts graduating later are not yet observed as independent adults. Third, I drop 30,320 respondents observed outside ages 22 to 36, of whom 3,309 are younger and 27,011 older. Fourth, I drop 33,087 respondents who live with a spouse or partner, because their banking status is shared with a partner, so a gender comparison among them reflects who was surveyed rather than their own banking decisions.

Because exposure is assigned by state and graduation cohort, it is variation across these cohorts and states, and not variation in age, that identifies the effect of the mandate. I set the two age bounds for separate reasons. I set the lower bound at 22 because of the household-level measurement noted above: respondents younger than 22 are often still enrolled in school and living with their parents, so their recorded status describes their parents’ banking rather than their own. I set the upper bound at 36 to hold the observation window comparable across cohorts, since the cohort restriction on its own would admit the earliest cohorts at much older ages. A respondent who graduated in 1991 is 50 by the 2023 wave, and the cap keeps every cohort observed over the same span of early adulthood.

Restricting the sample to unpartnered individuals minimizes the influence of a partner’s financial decisions on household banking status. I define unpartnered individuals as those living without a spouse or cohabiting partner, which counts respondents who are legally

married but whose spouse is absent from the household as unpartnered. I impose this restriction because among partnered respondents a gender comparison reflects which partner answered the survey rather than a difference in their own banking decisions. Banking status is recorded for the household rather than the individual, so a partnered woman and her partner are assigned the same status, and surveying either one returns the same household rate.

Treatment assignment. Treatment is defined as exposure to a state-mandated high school personal finance course. I take the timing of each state’s requirement, the first graduating cohort required to complete personal finance instruction, from Urban et al. (2020) and Urban’s compilation of state financial education requirements, last updated in 2024.⁴ The survey records the state a respondent lives in, not the state where they attended high school, so I assume the two coincide and that respondents graduated in the year they turned 18. Interstate migration between high school and the survey therefore assigns some respondents the wrong state’s requirement, and the estimates are intent-to-treat with respect to the assigned requirement rather than to coursework actually completed. I consider a respondent to have been treated if that state implemented a mandate no later than the year of their high school graduation. I use this any-exposure rule, rather than requiring that the mandate be in place before the respondent entered high school, because most students take the required course toward the end of high school, as described in Section 3.

Outcome and covariates. The primary outcome is whether the respondent lives in an unbanked household, which the FDIC defines as a household in which no member has a checking or savings account at a bank or credit union.

The FDIC supplement records respondents’ race and ethnicity, family income, education, marital status, age, labor force status, and whether they were born in the United States. I recover respondents’ gender and cohabitation status by linking the supplement to the

⁴Available at https://papers.carlyurban.com/Policies_ForPosting_2024.xlsx.

underlying CPS files. I combine marital status, cohabitation status, and the presence of children younger than eighteen to construct a measure of household structure. Educational attainment is classified into four categories: less than a high school diploma, high school graduate, some college, and college graduate. Family income is reported in five categories ranging from less than \$15,000 to \$75,000 or more. These are the survey’s own categories, recorded in nominal dollars and not adjusted for inflation, so income enters the analysis as a set of category indicators rather than as a real amount. Family income includes the income of all household members related to the reference person by birth, marriage, or adoption. Therefore, income from an unmarried partner or an absent spouse is excluded. Because relatively few respondents identify as Asian, I combine them with other racial and ethnic groups into an “Other” category, leaving indicators for White, Black, Hispanic, and Other.

I compare respondents in ever-treated and never-treated states in appendix Table A3. Respondents in ever-treated states are more likely to be unbanked and have lower incomes and less education. These differences motivate the empirical strategy, which compares cohorts exposed and unexposed to the mandate within the same state rather than comparing states with one another.

4.2 National Financial Capability Study

The NFCS is conducted every three years by the Financial Industry Regulatory Authority (FINRA) Investor Education Foundation, and I use all six waves, fielded from 2009 to 2024. It records measures of financial literacy, financial attitudes, and exposure to financial education, none of which the FDIC survey collects.

The NFCS differs from the FDIC survey in two ways that bear on what I can ask of it. It is an online, non-probability sample rather than a nationally representative household survey, and it records age in broad bands rather than in single years, which is what assigning respondents to graduation cohorts requires. For both reasons I use the NFCS to examine the financial knowledge mechanism, and I leave the effect of the mandate on banking to the

FDIC survey.

Exposure to financial education is recorded by a question asking whether a school, college, or workplace the respondent attended offered financial education, and whether the respondent took part. Respondents who did are then asked whether they received it in high school, in college, or from an employer. These items were first fielded in 2012, so exposure is measured in five of the six waves, while the financial knowledge questions are available in all six.

The primary outcome is financial knowledge, measured using the standard Big Five financial literacy questions developed by Lusardi and Mitchell (2011, 2014). Although a sixth question was introduced beginning in 2015, I use the original Big Five questions to maintain consistency across survey waves. I also examine self-assessed financial knowledge because studies have documented gender differences in financial confidence, even conditional on measured financial literacy (Bucher-Koenen et al., 2017; Aristei and Gallo, 2022).

To make the NFCS analysis comparable to the FDIC survey analysis, I restrict the sample to unpartnered high school graduates aged 18 to 44, which retains the youngest three of the six age bands the survey records (18–24, 25–34, and 35–44). Graduation cohorts can therefore be approximated only coarsely, and Section 6 sets out what that costs the design. The age floor of 18 raises no measurement problem here, because the NFCS records financial knowledge for the individual respondent rather than banking status for the household, so a respondent’s answers are their own. The covariates are more limited than in the FDIC survey, with race observed only as White versus non-White respondents. The estimates therefore control for gender and for non-White status, alongside state, graduation-cohort, and survey-year fixed effects.

I report the same comparison for the NFCS sample in appendix Table A4. Respondents in ever-treated states have somewhat lower measured financial knowledge and less education, although the two groups report similar rates of financial education. The NFCS therefore supplies what the FDIC survey cannot, a direct measure of what students learned, at the

cost of a coarser design.

5 Descriptive Patterns in the Gender Gap

The gender gap in being unbanked is concentrated among the groups least likely to hold an account at all, and locating where it is widest motivates the subgroup estimates that follow. In Table 1, I report the unbanked rate and the female–male gap across household structure, race and ethnicity, family income, education, and age. The gap is not uniform. It is widest among respondents with no education beyond high school, among Black and Hispanic respondents, and among those in the lowest income categories, and it is also large among never-married respondents with children. It does not vary with age, however: the gap is 5.84 percentage points at ages 22 to 26, 6.80 percentage points at ages 27 to 31, and 5.67 percentage points at ages 32 to 36, which weighs against reading it as a life-cycle phenomenon. In columns (5) through (7) of Table 5, I later estimate the effect of the mandate separately for these subgroups, and the estimated gender differentials are largest for education and for race and ethnicity.

I next ask whether women are more often found in the groups where the gap is widest, comparing the characteristics of women and men in the analysis sample in Table 2. As shown in the first row of Table 2, 13.28 percent of women lack a bank account, compared with 7.12 percent of men, so the rate among women is about 87 percent higher than among men. The two groups also differ along several dimensions. For example, women have lower family income and are substantially more likely to be single parents, whereas men in the sample are predominantly never-married without children. Women are also more likely to be out of the labor force and are more likely to be Black, which partly reflects the restriction to unpartnered respondents: 74.29 percent of Black women in this age and education window are unpartnered, compared with 40.17 percent of White women. At the same time, women are slightly more likely to have attended college, which would predict greater bank-

ing participation rather than a higher unbanked rate. With that one exception, these are the same groups in which the gap in Table 1 is widest: women are over-represented among never-married respondents with children, among Black respondents, and in the lowest income categories, while in education they are the better-educated group, so education works against the gap rather than toward it. These gender differences suggest that the raw gap partly reflects differences in economic circumstances and household structure, a possibility I quantify with the decomposition in Section 7.

Everything so far rests on characteristics the survey records. I turn now to what unbanked respondents say themselves, which bears on whether the gap reflects constraints women face or choices they make. In Table 3, I report the barriers they cite. In the first row, insufficient funds to meet minimum balance requirements is the most frequently cited barrier for both groups, and it is cited by 52.20 percent of unbanked women compared with 45.26 percent of unbanked men, a difference of 6.93 percentage points. The second row reverses the sign: 39.64 percent of men cite a lack of trust in banks, against 32.72 percent of women, a difference of 6.92 percentage points. These are the only two rows in which the gender difference is statistically significant. Women and men cite the remaining barriers at similar rates: fees (32.88 versus 37.01 percent), privacy (29.76 versus 31.93), identification (14.59 versus 17.45), product availability (14.93 versus 15.91), and inconvenient hours or locations (11.21 versus 14.44). The barrier women cite more often, not having enough money to meet a minimum balance, describes what they can afford, while the barrier men cite more often, distrust of banks, describes whether they want an account at all. Read that way, the gender gap in unbanked status is associated more with what women can afford than with whether they want to participate, which is also how the decomposition in Section 7 reads, since family income is the largest single contributor to the explained gap.

A natural question is whether unbanked women want a bank account less than unbanked men do. They do not. Among the 1,928 unbanked respondents observed in the 2009 through 2017 waves, which are the waves that carry the future-intentions questions, 45.75 percent

of women report wanting to open an account, compared with 37.65 percent of men, while 42.62 percent of men report having never held an account and having no plans to open one, compared with 29.72 percent of women. Willingness to participate in the formal banking system is therefore not the binding constraint for unbanked women. What this rules out is the preference channel: if unbanked women simply did not want accounts, a course that supplies information could not move their banking status. Money is one constraint that does bind, since insufficient funds to meet minimum balance requirements is the barrier they report most often, and a high school course cannot relax it. What such a course can relax is information: which accounts carry low or no minimum balances, and how to open and maintain one. I test that channel in Section 7.4. These are stated intentions, however, and I cannot rule out that young low-income men rationalize an outcome as a choice while women more readily report wanting something they lack. I therefore read this pattern as the *ex ante* reason to expect the mandate to reduce unbanked rates more among women than among men, which is what I estimate in Section 7, rather than as evidence on the causal channel itself.

These descriptive patterns share a common thread. The gender gap in being unbanked is concentrated among women with fewer economic resources and more demanding household circumstances, and these women more often report wanting a bank account than reporting no interest in one. Two questions follow, which I address in the remainder of the study: how much of the gap these differences in circumstances account for, and whether a policy that builds financial literacy can narrow it. The next section describes the two analyses I use to answer them.

6 Empirical Strategy

My empirical analysis has two parts, and each answers a different question. The first part is descriptive. I ask how much of the female–male gap in being unbanked comes from the

ways unpartnered women and men differ in income, education, household structure, and other characteristics. For this I use a Kitagawa–Oaxaca–Blinder decomposition, which splits the gap into the parts associated with each of these characteristics but does not identify the effect of changing any of them. The second part is causal. I ask whether a specific policy, a state requirement to take personal finance in high school, lowers the unbanked rate, and whether it lowers it more for women than for men and so narrows the gap. For this I use a staggered difference-in-differences design based on the extended two-way fixed-effects estimator of Wooldridge (2025).

I begin with the decomposition (Kitagawa, 1955; Oaxaca, 1973; Blinder, 1973), which is based on a linear probability model.⁵ It separates the gender gap in unbanked status into an explained component arising from differences in observed characteristics and an unexplained component, which is what remains when women and men with the same characteristics still differ in how likely they are to be unbanked. I include income, education, labor-force status, household structure, race and ethnicity, whether the respondent was born in the United States, and age as covariates. I estimate the explained component using coefficients from a linear probability model fit to the pooled sample of women and men, following the pooled-model variant of Fortin, Lemieux and Firpo (2011). The pooled model includes an indicator for gender, which prevents the pooled coefficients from absorbing part of the gender differential and reassigning it from the unexplained component to the explained component. It measures how much of the gap would close if women held the same characteristics as men, with each characteristic valued at a common association with being unbanked. Each characteristic contributes the difference in its mean between women and men, multiplied by that common coefficient, so a characteristic accounts for more of the gap when women and men differ on it and when it is more strongly related to being unbanked. The unexplained

⁵A nonlinear decomposition approach for binary outcomes, such as the method proposed by Fairlie (2005), is also a natural choice in this setting. However, the estimates from that approach were unstable in my setting with the full set of categorical controls. The simulation-based procedure and state-clustered bootstrap inference generated large variation, likely because some combinations of covariates contained relatively few observations. I therefore use the linear probability model as an alternative.

component is the remainder, and it reflects differences in the estimated relationship between characteristics and banking status together with characteristics the FDIC survey data do not measure.

The second part of the analysis estimates the causal effect of the mandate. The decomposition takes the characteristics that differ between women and men as given. What I want here is the effect of the mandate itself on the probability of being unbanked, estimated separately for women and for men, together with the difference between those two effects, which is the change in the gender gap. Estimating it requires variation in exposure that is not itself a characteristic of the respondent, and the staggered rollout of the mandates supplies exactly that: states adopted personal finance requirements at different times, so respondents who graduated in the same year faced different requirements depending on the state they were schooled in, and respondents in the same state faced different requirements depending on the year they graduated. Comparing exposed with unexposed cohorts along both margins is a staggered difference-in-differences design, and I implement it with the extended two-way fixed-effects (ETWFE) estimator of Wooldridge (2025).

In Figure 1, I plot the identifying variation: the staggered arrival of the requirement across graduating cohorts, and the share of the sample exposed in each cohort bin. The estimator has to accommodate the fact that states adopted in different years, because the effect of the mandate can differ across adoption cohorts. ETWFE does so by estimating a separate treatment effect for every combination of adoption cohort and graduation cohort, drawing comparisons only against cohorts that are not yet treated or never treated, and then averaging those effects into a single summary. This is what recommends it over a conventional two-way fixed-effects (TWFE) regression carrying a single treatment indicator, which in a staggered setting uses already-treated cohorts as controls for later-treated ones. Such forbidden comparisons can receive negative weight in the estimated average effect (Goodman-Bacon, 2021). The Callaway–Sant’Anna and Sun–Abraham estimators avoid them as well, and I report both alongside ETWFE in the event-study figure. I prefer ETWFE for the

main estimates because it recovers all cohort-specific effects within a single pooled regression, which allows the difference between women’s and men’s effects, and the standard error of that difference, to come from one clustered covariance matrix. Callaway–Sant’Anna estimates group-time effects separately, and Sun–Abraham is designed to characterize the dynamic path of an effect rather than a contrast between two groups. Since the object of interest here is that contrast, ETWFE is the natural choice. A Goodman-Bacon decomposition of the FDIC survey sample indicates that forbidden comparisons carry only a small share of the weight in this setting, which is why conventional TWFE suffices for the NFCS analysis, where the smaller sample and the coarser cohort measure make the staggered-adoption estimators less feasible.

The mechanics are as follows. Let $s(i)$ denote respondent i ’s state and $c(i)$ their high school graduating cohort, defined as the year they turned 18. State s first requires the mandate for cohort g_s , with $g_s = \infty$ for never-treated states, so respondent i is exposed if $c(i) \geq g_{s(i)}$. Eleven states adopted mandates that first apply to cohorts graduating in 2020 or later, after the last cohort I observe, so respondents in those states are never exposed within the sample and serve as not-yet-treated controls. Let $D_i^{gt} = \mathbf{1}\{g_{s(i)} = g, c(i) = t, t \geq g\}$ indicate that respondent i belongs to treated cell (g, t) , so that each treated respondent has $D_i^{gt} = 1$ for one cell only and untreated respondents have $D_i^{gt} = 0$ for every cell. I estimate

$$Y_i = \alpha_{g_{s(i)}} + \lambda_{c(i)} + X_i' \beta + \gamma \text{female}_i + \sum_g \sum_{t \geq g} (\tau_{gt} + \delta_{gt} \text{female}_i) D_i^{gt} + \varepsilon_i \quad (1)$$

by OLS, with standard errors clustered at the state level. Here Y_i is an indicator for being unbanked, α_g are adoption-cohort fixed effects, λ_t are graduation-cohort fixed effects, female_i is an indicator equal to one if respondent i is female, and X_i contains indicators for Black, Hispanic, and other race or ethnicity, with White the omitted category, together with an indicator for having been born outside the United States. These are the characteristics fixed before high school, and I return below to why the richer covariate list used in the

decomposition does not belong here. The two sets of fixed effects absorb differences across states that adopted at different times and differences across cohorts entering adulthood at different times, which is what lets the design apply to the FDIC survey even though it is a repeated cross-section rather than a panel. To avoid sparse treatment cells, graduation cohorts are grouped into three-year bins.

Three quantities follow from this equation. Within a treated cell (g, t) , τ_{gt} is the effect of the mandate on men, $\tau_{gt} + \delta_{gt}$ is the effect on women, and δ_{gt} is the difference between them, which is the change in the gender gap. Because there is one such set of parameters for every treated cell, I summarize them as averages over the treated. The average treatment effect on the treated is a count-share-weighted average of the cell-specific effects, each cell weighted by its share of treated respondents, and the female and male averages use the corresponding within-gender shares. These averages are what I report in Table 5: the Mandate exposure row is the average of τ_{gt} , the effect for men, the Mandate exposure $\times$ Female row is the average of δ_{gt} , the change in the gap, and the effect for women is the sum of the two. Column (1) gives the same model with δ_{gt} restricted to zero and no covariates, which yields one average effect for the sample as a whole.

Unlike Wooldridge (2025), who allows control variables to interact with each treatment cell and period, I include control variables additively. This specification trades some flexibility in the conditional parallel-trends assumption for greater precision. Identification requires that, conditional on controls and fixed effects, treated and untreated cohorts would have experienced similar changes in banking participation in the absence of the mandate. I assess this assumption using event-study estimates, which compare outcomes across cohorts before and after mandate exposure. The coefficients on pre-treatment periods capture whether treated cohorts experienced differential changes in unbanked rates prior to the mandate. The event-study estimates in Figure 2 give no evidence of differential pre-treatment trends, and joint tests fail to reject the null of zero pre-treatment effects for women, men, and the female–male gap. The identifying assumption also requires no anticipation of treatment,

which is plausible because the policy affects students through high school coursework and cannot directly change adult banking outcomes before exposure.

A final consideration is that the two parts of the analysis use different sets of covariates, and this difference is deliberate. In the difference-in-differences design I control only for characteristics that are fixed before high school, namely race, ethnicity, and whether the respondent was born in the United States. I do not control for income, education, or employment, because the mandate can itself affect these outcomes, and conditioning on variables that the treatment can change would absorb part of the effect I am trying to measure. Adding them attenuates the estimated gender differential, as I report in appendix Table A5. I therefore study income, education, and employment as outcomes in Section 7.4, asking whether the mandate improves them, rather than holding them fixed here. This concern does not arise in the decomposition, which is descriptive and estimates no treatment effect, so there I include income, education, and employment alongside the other characteristics.

Column (4) of Table 5 is therefore my preferred specification. It is the one that estimates the difference between women’s and men’s effects, which is the parameter this study is about, and it conditions on the full set of pre-high-school characteristics while conditioning on nothing the mandate could itself have changed. Columns (2) and (3) reach it by adding those characteristics in steps, and I report the build-up because the estimated difference barely moves as they enter, which is itself evidence that the difference is not an artifact of composition.

7 Findings

I report the decomposition first, and then the difference-in-differences estimates of the mandate’s effect on the gap.

7.1 Decomposing the Gap

In Table 4, I report the decomposition. Differences in observed characteristics account for 5.05 of the 6.20 percentage-point gap, or approximately 81 percent. Differences in income and household structure account for most of the explained gap. Income differences contribute about one third of the overall gap, while differences in household structure, particularly women’s higher likelihood of being single parents, contribute roughly 28 percent. Race and ethnicity explain a smaller share, while differences in education offset part of the gap. The unexplained component accounts for the remaining 1.15 percentage points, or roughly one fifth of the gap.⁶ In appendix Table A1, I report the sample means by gender and the estimated coefficients that underlie each contribution. Economic circumstances, then, account for most of the gap. Whether a policy can narrow it is a causal question, and I turn to it now.

7.2 Mandates Reduce the Gender Gap in Being Unbanked

In Table 5, I report the effect of high school personal finance mandates on the probability of being unbanked among unpartnered young adults. Column (1) gives the estimated overall average treatment effect with no additional covariates: mandate exposure is predicted to reduce the probability of being unbanked by 2.54 percentage points, roughly one quarter of the 10.33 percent baseline unbanked rate shown in the Mean dep. var. row.

In columns (2)–(4), I ask whether this effect differs by gender by interacting mandate exposure with an indicator for female. Column (2) includes only the interaction term, while columns (3) and (4) progressively add demographic controls. The reduction in being unbanked proves much larger for women than for men. In the preferred specification (col-

⁶In a further check that I do not report, I add to the decomposition an indicator for whether the respondent was exposed to a personal finance mandate in high school. Mandate exposure explains essentially none of the gap (an estimated 0.00 percentage points), because exposure is assigned by state and graduation cohort and is balanced between women and men. The decomposition measures how much of the gap reflects characteristics on which women and men differ, and the mandate is not one of them. Its effect is identified separately by the difference-in-differences design.

umn 4), mandate exposure is predicted to lower men’s probability of being unbanked by 1.23 percentage points, or about 17 percent of the 7.12 percent rate among men. Women are predicted to experience a further 2.53 percentage point decline, for a total reduction of 3.76 percentage points, or about 28 percent of the 13.28 percent rate among women. The estimated effect for women is roughly three times the effect for men. The estimated gender difference remains stable across specifications, changing little as demographic controls are added, suggesting that it is not explained by differences in race, ethnicity, or whether respondents were born in the United States. Relative to the 6.20 percentage point female–male gap reported in Table 4, the additional 2.53 percentage point reduction for women implies that mandate exposure closes roughly 41 percent of the gender gap.

In Figure 2, I report these estimates by graduation cohort relative to the first cohort the requirement binds on, separately for women and men. Women’s probability of being unbanked falls as soon as the requirement binds, and the estimates for the first two exposed cohort bins are distinguishable from zero. They remain negative at later horizons, where fewer exposed cohorts are observed and the estimates are less precise. Men’s estimates are not distinguishable from zero at any horizon. The pre-treatment coefficients, which Section 6 uses to assess the identifying assumption, are close to zero for both groups.⁷

7.3 Which Women Benefit Most from the Mandates?

The estimates so far describe the average unpartnered woman. If the mandate works by supplying information that women would not otherwise get, its effect should be largest among those least likely to encounter that information elsewhere. I therefore re-estimate Equation (1) separately on three subgroups that proxy for disadvantage: Black or Hispanic respondents, foreign-born respondents, and respondents with no education beyond

⁷Because the literature offers several heterogeneity-robust estimators for staggered adoption, I re-estimate the event study using Callaway–Sant’Anna, Sun–Abraham, and dynamic TWFE, and report the four side by side in appendix Figure A1. The estimates are broadly similar, so the result is not an artifact of the estimator.

high school.⁸

Columns (5) through (7) of Table 5 repeat the column (4) specification on these three subsamples, so each is read against the full-sample estimate of -2.53 in column (4). Among Black and Hispanic respondents (column 5), the estimated differential is -3.79 percentage points on a base of 18.66 percent, about 20 percent of that group’s unbanked rate. Among foreign-born respondents (column 6), the differential is -0.59 percentage points and is not statistically distinguishable from zero; this group starts from the lowest baseline unbanked rate, 9.80 percent, and the female–male gap within it is itself close to zero, as the 1 if female coefficient in column (6) shows, so little gender gap remains for the mandate to close. Among respondents with no education beyond high school (column 7), the differential is -7.32 percentage points on a base of 22.71 percent, about 32 percent, the largest reduction of any subgroup.

The reductions concentrate where baseline exclusion is highest. Respondents with no education beyond high school and Black and Hispanic respondents start from unbanked rates of 22.71 and 18.66 percent, well above the 10.33 percent full-sample rate, and it is in these two groups that the mandate does the most. Foreign-born respondents, whose baseline rate of 9.80 percent is close to the full-sample level, show essentially no differential. The requirement therefore does most where exclusion is widest at baseline and where its content is least likely to be encountered again.

7.4 Mechanisms: Why the Mandates Help Women More

Several mechanisms could produce a larger effect for women, and I examine two of them. The mandate could raise what women know about finding and using an account, which is what the course is designed to do. It could also change the circumstances that make an

⁸Students from low-income households are a natural group to examine, since they may gain the most from a personal finance course, which is not to say that students from higher-income households gain nothing. I do not split the sample on income, however, because the family income the survey records is contemporaneous rather than measured at high school age, and it enters as an outcome in the mechanism analysis of Section 7.4. I instead use race and ethnicity, nativity, and education as proxies for disadvantage.

account worth holding, by raising education, shifting the kind of work women do, or raising income. The FDIC survey can test the second, because it records education, employment, and income, but not the first, because it asks nothing about financial knowledge. I therefore take the economic channel first, in the FDIC survey, and then turn to the knowledge channel in the NFCS.

7.4.1 Economic and Human Capital Outcomes

In Table 6, I examine whether mandates affect economic and human capital outcomes closely related to financial inclusion, to better understand why they reduce unbanked rates more among women than among men. I estimate Equation (1), the specification that allows the effect of the mandate to differ between women and men, with the same adoption-cohort and graduation-cohort fixed effects, the same controls for race, ethnicity, and whether the respondent was born in the United States, and the same state-clustered standard errors. Only the outcome changes. In place of being unbanked I use college completion, labor-force participation and employment, self-employment overall and split into its incorporated and unincorporated forms, and an indicator for each family-income category.

Columns (1) and (2) of Table 6 give the estimates for educational attainment. College completion is an indicator for holding a bachelor's degree or higher, so an associate degree does not count toward it, while the some college category covers respondents who attended without completing a degree together with those holding an associate degree. The two categories are mutually exclusive, with high school only as the residual, so the estimates describe how the mandate shifts respondents across an exhaustive set of attainment levels. In column (2), women are predicted to complete college 2.71 percentage points more often than men, an increase of about 7 percent of the base rate of 41.37 percent. The estimated effect for men is 0.90 percentage points and is not statistically distinguishable from zero. In column (1), the differential for some college is -0.56 percentage points on a mean of 33.15 percent, precisely estimated around zero.

Columns (3) through (5) give the labor force outcomes, and none of the three differentials is statistically significant. The point estimates are small relative to their means: 1.14 percentage points on employment (mean 82.59 percent), 0.09 on unemployment (mean 6.03), and -1.23 on being out of the labor force (mean 11.38). Their standard errors are 1.28, 0.42 and 1.02, so effects larger than about 2.6, 0.8 and 2.0 percentage points in either direction lie more than two standard errors from these estimates. These are therefore informative nulls rather than merely imprecise ones. Columns (6) through (8) are estimated among employed respondents only, so they describe the kind of work people do rather than whether they work. In column (6), women are predicted to increase their probability of self-employment by 1.67 percentage points more than men, an increase of about a third of the 5.12 percent base rate, while the estimate for men, 0.78 percentage points, is not statistically significant. The gain is concentrated in unincorporated self-employment, where the differential in column (8) is 1.33 percentage points on a mean of 3.58 percent. Incorporated self-employment moves the other way. In column (7), the differential is 0.34 percentage points and not significant, whereas men are predicted to increase incorporated self-employment by 1.06 percentage points from a mean of 1.54 percent.

Columns (9) through (12) give the four mutually exclusive family-income categories. Because family income includes the income of related household members, in this unpartnered sample it is close to but not the same as the respondent's own earnings. Men are predicted to move out of the lowest category, by -2.96 percentage points on a mean of 37.85 percent in column (9), and into the highest, by 3.79 percentage points on a mean of 19.23 percent in column (12). Women's gains sit in the middle instead: in column (11), the differential is 3.85 percentage points in the \$50,000 to \$75,000 category, whose mean is 18.72 percent, with a differential of -2.45 percentage points at \$75,000 or more in column (12).

The economic mechanism is therefore visible in the data. Relative to men, exposed women complete college more often, enter unincorporated self-employment more often, and more often reach the middle of the income distribution, and each of these changes makes a

bank account more valuable to hold.

7.4.2 Financial Knowledge

The second mechanism is the one the course targets directly: what students know. Financial knowledge appears nowhere in the FDIC survey data, so the decomposition could not account for it and the estimates so far cannot detect it. I therefore turn to the NFCS, which records both exposure to financial education and financial knowledge, and ask whether the mandate raises each, for women and for men.

As described in Section 4, the sample is unpartnered high school graduates ages 18 to 44, and the coarse age bands leave TWFE models, with state, graduation-cohort, and survey-year fixed effects and standard errors clustered at the state level, as the feasible design.

I first examine whether mandates increase exposure to financial education, reporting the estimates in Table 7. In column (2), mandate exposure is predicted to raise the probability of reporting any financial education by 2.09 percentage points from a mean of 24.08 percent, and in column (3) to raise financial education received in high school by 2.57 percentage points from a mean of 14.85 percent. The gender differentials in the top row, 0.47 and 0.29 percentage points, are not statistically significant, so the mandate is estimated to reach women and men alike.

In Table 8, I then examine whether mandates translate into improvements in financial knowledge. Unlike exposure to financial education, the estimated effects on knowledge are larger among women. In column (1), the estimated gender difference in the Big Five score is 0.13 points on a mean of 2.22, about 6 percent of the mean, while the estimate for men is -0.05 and not significant. The gain sits in the two investment questions: in columns (4) and (6), the differentials are 4.35 percentage points on bond prices (mean 20.32 percent) and 4.28 on stock-versus-fund risk (mean 35.07 percent), gains of about a fifth and about an eighth of their respective base rates. Column (2) gives a smaller differential of 2.61 percentage points on compound interest, where the mean is 65.20 percent. In column (7),

self-rated knowledge does not move, with a differential of -0.03 points on a mean of 4.80. The mandate therefore delivers financial education to women and men alike but produces larger gains in women’s measured knowledge, and what improves is what respondents know rather than their confidence in it.

8 Conclusion

Among unpartnered young adults, whose recorded banking status is their own rather than a partner’s, women are substantially more likely than men to be unbanked. In this study I ask whether a high school personal finance requirement narrows that gap, and whether it does so directly, by raising what young women know, or indirectly, by improving the economic circumstances that make an account worth holding. Using the FDIC National Survey of Unbanked and Underbanked Households and a heterogeneity-robust difference-in-differences design, I find that exposure to a requirement reduces the probability of being unbanked by about 2.54 percentage points overall, equivalent to roughly one quarter of the baseline unbanked rate, and that the gain is concentrated among women: the effect for women is roughly three times the effect for men and closes about 41 percent of the gender gap among unpartnered young adults.

The effects are largest among women who face the greatest barriers to financial inclusion at baseline, namely women with no education beyond high school and Black and Hispanic women. Both channels are visible, and both are modest. Financial knowledge rises. Relative to men, exposed women score 0.13 points higher on the five-question scale, about 6 percent of the mean, with the gain concentrated in the two investment questions, and their self-assessed knowledge does not move. Economic circumstances improve as well, with college completion rising by 2.71 percentage points relative to men, unincorporated self-employment by 1.33 points, and the probability of holding income between \$50,000 and \$75,000 by 3.85 points. The two channels are estimated on different samples and different surveys, so the

design cannot apportion the effect on banking between them.

These findings contribute to the broader discussion of how to improve financial inclusion. Many adult financial education programs (Fernandes, Lynch and Netemeyer, 2014) and efforts to expand account access (Dupas et al., 2018; Collins, Larrimore and Urban, 2026) have had limited effects. The constraint is not that adults cannot learn what such a course teaches. It is that these programs depend on people enrolling in them, so they tend to reach those already engaged with the financial system and to miss those who are not. A graduation requirement removes that choice: it reaches every student in an affected cohort, including those who would not have sought financial education on their own, and it reaches them before they begin managing money independently. The estimates here also suggest that account access is not the binding constraint for these women. Unbanked women report wanting an account more often than unbanked men do, and what moves when the requirement binds is their measured knowledge, not their stated willingness. Where the barrier is informational, expanding the supply of accounts is unlikely to be sufficient on its own.

Three questions follow. The first is why the same course is more useful to young women than to young men, which these estimates establish but do not explain, and which bears directly on whether financial education should be delivered more widely or designed differently. The second is whether the gains compound: banking status is observed only through the 2019 graduating cohort, which leaves the longer-run consequences for saving, credit, and business ownership open. The third is how far these results travel. The design requires respondents whose banking status is their own, so it is silent on adults who are already unbanked and on people who make banking decisions jointly with a partner, and neither group should be assumed to respond as this sample does.

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Tables

Table 1: Unbanked rates by gender and respondent characteristics

Characteristic	Female (1)	Male (2)	Difference (3)	N (F) (4)	N (M) (5)
All unpartnered respondents	13.28 (0.66)	7.12 (0.50)	+6.16***	13,442	11,962
Household structure					
never married, with children	26.13 (1.09)	15.35 (2.11)	+10.78***	3,536	588
never married, no children	5.70 (0.44)	6.16 (0.48)	−0.46	6,717	9,638
formerly married, with children	15.31 (1.00)	8.80 (1.51)	+6.52***	2,045	374
formerly married, no children	9.51 (1.50)	10.05 (1.52)	−0.53	717	1,066
married (spouse absent), with children	20.27 (2.94)	7.62 (4.44)	+12.65***	260	28
married (spouse absent), no children	13.25 (4.07)	8.98 (1.45)	+4.27	167	268
Race and ethnicity					
White	6.77 (0.62)	3.81 (0.30)	+2.96***	7,558	8,016
Black	25.50 (1.37)	16.98 (1.23)	+8.52***	2,854	1,268
Hispanic	15.53 (1.63)	11.36 (1.04)	+4.17**	1,829	1,432
other race or ethnicity	7.70 (1.67)	3.69 (0.66)	+4.01***	1,201	1,246
Family income					
under \$15,000	35.13 (1.26)	23.87 (1.71)	+11.26***	2,961	1,540
\$15,000 to \$30,000	17.15 (0.98)	12.32 (1.08)	+4.83***	3,028	1,962
\$30,000 to \$50,000	4.63 (0.44)	4.59 (0.44)	+0.05	3,246	3,016
\$50,000 to \$75,000	2.77 (0.44)	2.29 (0.36)	+0.48	2,085	2,503
\$75,000 or more	1.39 (0.36)	1.18 (0.34)	+0.20	1,893	2,721
Education					
high school graduate	28.96 (1.26)	16.73 (1.19)	+12.23***	3,210	3,259
some college	15.56 (1.00)	6.74 (0.63)	+8.82***	4,648	3,706
college or more	2.14 (0.31)	1.16 (0.15)	+0.98***	5,584	4,997
Age					
22 to 26	12.48 (0.88)	6.64 (0.61)	+5.84***	4,132	3,909
27 to 31	13.58 (0.87)	6.78 (0.61)	+6.80***	4,804	4,331
32 to 36	13.74 (0.78)	8.07 (0.68)	+5.67***	4,506	3,722

Note: I report the survey-weighted share unbanked, by gender, in the unpartnered sample. Household structure combines marital history with the presence of a co-resident child aged 18 or younger; married respondents are spouse-absent, since respondents with a co-resident spouse or partner are excluded by construction, and formerly married covers divorced, separated, and widowed respondents. The Difference column is the female share minus the male share, in percentage points; the spouse-absent rows contain few male respondents, so their differences are imprecisely estimated. Standard errors clustered at the state level are in parentheses. Stars are from a weighted, state-clustered t -test of the difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Sample means of banking status and covariates by gender

Variable	Total (1)	Female (2)	Male (3)	Difference (4)
Banking status				
1 if unbanked	10.33 (0.54)	13.28 (0.66)	7.12 (0.50)	+6.16***
Household structure				
1 if never married, with children	17.21 (0.55)	28.07 (0.92)	5.36 (0.48)	+22.72***
1 if never married, no children	65.02 (0.94)	50.38 (1.25)	81.00 (0.73)	-30.62***
1 if formerly married, with children	8.36 (0.45)	13.54 (0.71)	2.71 (0.23)	+10.83***
1 if formerly married, no children	6.61 (0.30)	5.08 (0.27)	8.28 (0.49)	-3.19***
1 if married (spouse absent), with children	1.07 (0.06)	1.82 (0.11)	0.25 (0.05)	+1.57***
1 if married (spouse absent), no children	1.72 (0.14)	1.10 (0.14)	2.40 (0.18)	-1.30***
Race and ethnicity				
1 if White	52.05 (2.66)	47.89 (2.37)	56.58 (3.08)	-8.68***
1 if Black	21.62 (2.21)	26.70 (2.63)	16.07 (1.70)	+10.63***
1 if Hispanic	16.09 (2.74)	16.23 (2.81)	15.94 (2.70)	+0.29
1 if other race or ethnicity	10.24 (1.33)	9.17 (1.18)	11.41 (1.52)	-2.24***
Family income				
1 if family income is under \$15,000	17.61 (0.70)	21.74 (0.97)	13.10 (0.51)	+8.65***
1 if family income is \$15,000 to \$30,000	19.35 (0.75)	22.20 (0.75)	16.23 (0.89)	+5.97***
1 if family income is \$30,000 to \$50,000	24.55 (0.68)	24.35 (0.67)	24.77 (0.82)	-0.42
1 if family income is \$50,000 to \$75,000	18.99 (0.42)	16.71 (0.51)	21.48 (0.54)	-4.77***
1 if family income is \$75,000 or more	19.51 (1.63)	15.00 (1.45)	24.43 (1.86)	-9.42***
Education				
1 if high school graduate	25.49 (0.97)	23.90 (0.94)	27.22 (1.20)	-3.31***
1 if some college	33.15 (0.98)	35.24 (1.16)	30.86 (0.96)	+4.38***
1 if college or more	41.37 (1.54)	40.86 (1.69)	41.92 (1.56)	-1.06
Labor force participation				
1 if employed	82.59 (0.57)	80.43 (0.63)	84.94 (0.68)	-4.50***
1 if unemployed	6.03 (0.27)	6.68 (0.30)	5.33 (0.31)	+1.35***
1 if not in the labor force	11.38 (0.40)	12.89 (0.48)	9.73 (0.49)	+3.16***
Foreign-born (1 if born outside the U.S.)	13.47 (1.52)	11.74 (1.50)	15.37 (1.56)	-3.63***
Age in years	28.90 (0.05)	29.03 (0.06)	28.77 (0.06)	+0.26***
Observations	25,404	13,442	11,962	

Note: I report survey-weighted means of respondent characteristics, by gender. Household structure combines partnership status with the presence of a co-resident child aged 18 or younger; married respondents are spouse-absent, since respondents with a co-resident spouse or partner are excluded by construction, and formerly married covers divorced, separated, and widowed respondents. The Difference column is the female mean minus the male mean. Standard errors clustered at the state level are in parentheses. Stars are from a weighted, state-clustered t -test of the difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Reasons for being unbanked, by gender (2013–2023)

Reason	Total	Female	Male	Difference
	(1)	(2)	(3)	(4)
Don't have enough money to meet minimum balance	49.92	52.20	45.26	+6.93**
Don't trust banks	34.99	32.72	39.64	−6.92**
Bank account fees too high or unpredictable	34.24	32.88	37.01	−4.13
Avoiding a bank gives more privacy	30.47	29.76	31.93	−2.17
Don't have required personal identification	15.53	14.59	17.45	−2.86
Banks don't offer needed products or services	15.25	14.93	15.91	−0.98
Inconvenient bank hours or locations	12.27	11.21	14.44	−3.23
Observations (unbanked)	1,559	1,072	487	

Note: I report the survey-weighted share of unbanked respondents citing each reason, by gender. Respondents can report more than one reason, so the shares do not sum to 100. The Difference column is the female share minus the male share, in percentage points. Stars are from a weighted, state-clustered t -test of the difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Kitagawa–Oaxaca–Blinder decomposition of the gender gap in unbanked status

	Contribution	Share
	(1)	(2)
Total gap (Female – Male)	+6.20	100%
	(0.63)	
Explained	+5.05	+81%
	(0.42)	
Family income	+1.98	+32%
	(0.15)	
Household structure	+1.75	+28%
	(0.22)	
Race and ethnicity	+1.06	+17%
	(0.12)	
Labor force participation	+0.42	+7%
	(0.07)	
Age	+0.06	+1%
	(0.01)	
Foreign-born	+0.01	+0%
	(0.02)	
Education	–0.23	–4%
	(0.10)	
Unexplained	+1.15	+19%
	(0.40)	

Note: The decomposition is based on a linear probability model; the explained component prices the female–male difference in characteristics with coefficients from the pooled sample. Contributions are in percentage points and block contributions sum to the explained total by construction; column (2) gives each row’s contribution as a share of the total gap. The sample is 24,955 respondents (13,213 women and 11,742 men), and the underlying means and coefficients are in Table A1. Standard errors, below each estimate, are from 400 state-clustered bootstrap resamples.

Table 5: ETWFE estimates of the effect of high school personal finance mandates on the probability of being unbanked

	Dependent variable: 1 if unbanked						
	Full sample				Subgroups		
	(1)	(2)	(3)	(4)	Black/Hisp	Foreign born	HS only
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Mandate exposure $\times$ Female	—	−2.65*** (0.78)	−2.54*** (0.68)	−2.53*** (0.68)	−3.79** (1.82)	−0.59 (1.77)	−7.32*** (1.77)
Mandate exposure	−2.54*** (0.74)	−1.32* (0.78)	−1.24* (0.70)	−1.23* (0.70)	−2.49 (1.80)	−6.66*** (2.14)	−0.64 (1.60)
1 if female	—	6.60*** (0.62)	4.63*** (0.52)	4.59*** (0.51)	7.11*** (1.09)	−0.78 (1.03)	10.36*** (1.34)
1 if Black	—	—	17.74*** (1.15)	17.82*** (1.15)	7.78*** (1.48)	3.68* (1.92)	24.94*** (1.76)
1 if Hispanic	—	—	9.82*** (0.93)	10.18*** (0.92)	—	18.27*** (2.00)	13.27*** (1.55)
1 if other race or ethnicity	—	—	2.46** (1.07)	2.98** (1.19)	—	−0.61 (1.24)	8.75*** (2.52)
1 if foreign-born	—	—	—	−1.42** (0.68)	1.58 (1.06)	—	1.72 (1.88)
Mean dep. var. (%)	10.33	10.33	10.33	10.33	18.66	9.80	22.71
Observations	25,404	25,404	25,404	25,404	7,383	2,876	6,469
Adjusted R^2	0.007	0.018	0.066	0.066	0.028	0.099	0.093

Note: All columns are ETWFE estimates with adoption-cohort and graduation-cohort fixed effects, and coefficients are in percentage points. Columns (5)–(7) restrict the sample to Black or Hispanic respondents, foreign-born respondents, and respondents with no education beyond high school. Standard errors clustered at the state level are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: ETWFE estimates of the effect of high school personal finance mandates on human capital and economic outcomes

	Education		Labor force status			Self-employment among employed			Income			
	Some college	College+	Employed	Unemployed	Not in LF	Self-emp.	SE inc.	SE uninc.	< 30k	30-50k	50-75k	75k+
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Mandate exposure $\times$ Female	-0.56 (1.08)	2.71** (1.08)	1.14 (1.28)	0.09 (0.42)	-1.23 (1.02)	1.67*** (0.48)	0.34 (0.29)	1.33*** (0.44)	-0.69 (1.53)	-0.72 (1.03)	3.85*** (1.02)	-2.45** (1.01)
Mandate exposure	-0.64 (1.50)	0.90 (2.36)	-0.56 (1.16)	0.51 (0.67)	0.05 (1.09)	0.78 (0.70)	1.06*** (0.36)	-0.28 (0.68)	-2.96** (1.26)	-0.93 (1.23)	0.10 (1.36)	3.79*** (1.35)
1 if female	2.60*** (0.82)	2.19** (0.88)	-4.19*** (0.68)	0.38 (0.30)	3.81*** (0.58)	-3.24*** (0.28)	-1.46*** (0.16)	-1.78*** (0.24)	13.16*** (1.18)	-0.92 (0.74)	-5.44*** (0.80)	-6.80*** (0.70)
1 if Black	6.73*** (1.16)	-19.63*** (1.46)	-10.20*** (1.07)	5.93*** (0.48)	4.26*** (0.79)	-1.34*** (0.42)	-0.51** (0.22)	-0.84** (0.39)	16.48*** (1.50)	-2.58*** (0.96)	-5.90*** (0.84)	-8.00*** (1.12)
1 if Hispanic	9.02*** (1.46)	-24.62*** (1.71)	-2.49** (1.04)	1.82*** (0.49)	0.66 (0.78)	-0.72 (0.63)	-0.44* (0.24)	-0.29 (0.52)	8.75*** (1.48)	0.48 (0.97)	-3.12*** (1.02)	-6.11*** (1.70)
1 if other race or ethnicity	-1.04 (1.68)	4.84 (3.03)	-6.67*** (1.35)	1.17* (0.70)	5.49*** (1.01)	-0.27 (0.64)	-0.43 (0.28)	0.16 (0.55)	2.12 (1.91)	-6.30*** (1.09)	-2.18*** (0.83)	6.35*** (1.88)
1 if foreign-born	-10.59*** (0.93)	13.37*** (2.02)	0.47 (1.05)	-2.28*** (0.52)	1.81** (0.80)	0.51 (0.83)	0.13 (0.34)	0.38 (0.73)	-1.03 (1.25)	-2.09** (1.02)	0.64 (1.22)	2.48*** (0.86)
Mean dep. var. (%)	33.15	41.37	82.59	6.03	11.38	5.12	1.54	3.58	37.85	24.20	18.72	19.23
Observations	25,404	25,404	25,404	25,404	25,404	21,129	21,129	21,129	25,404	25,404	25,404	25,404
Adjusted R^2	0.019	0.062	0.018	0.012	0.010	0.007	0.003	0.003	0.049	0.006	0.009	0.044

Note: Each column is a separate ETWFE regression of a binary outcome, with adoption-cohort and graduation-cohort fixed effects, and coefficients are in percentage points. Self-employment outcomes are measured among employed respondents only. The income categories are mutually exclusive: under \$30k, \$30–50k, \$50–75k, and \$75k or more. Standard errors clustered at the state level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: TWFE estimates of the effect of high school personal finance mandates on the receipt of financial education

	Offered	Received	Received in HS
	(1)	(2)	(3)
Mandate exposure $\times$ Female	0.54 (1.26)	0.47 (0.92)	0.29 (0.74)
Mandate exposure	0.34 (1.42)	2.09* (1.16)	2.57** (1.10)
1 if female	-9.10*** (0.76)	-4.57*** (0.48)	-3.97*** (0.56)
1 if non-White	3.12*** (0.85)	2.12*** (0.67)	0.15 (0.45)
Mean dep. var.	37.86	24.08	14.85
Observations	26,421	26,421	26,421
Adjusted R^2	0.037	0.022	0.030

Note: All columns are static TWFE estimates with state, graduation-cohort, and survey-year fixed effects, on the NFCS sample of unpartnered respondents, and coefficients are in percentage points. ‘Received’ is financial education received in school, college, or at work and ‘Received in HS’ in high school, available only in NFCS waves from 2012 onward. Standard errors clustered at the state level are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

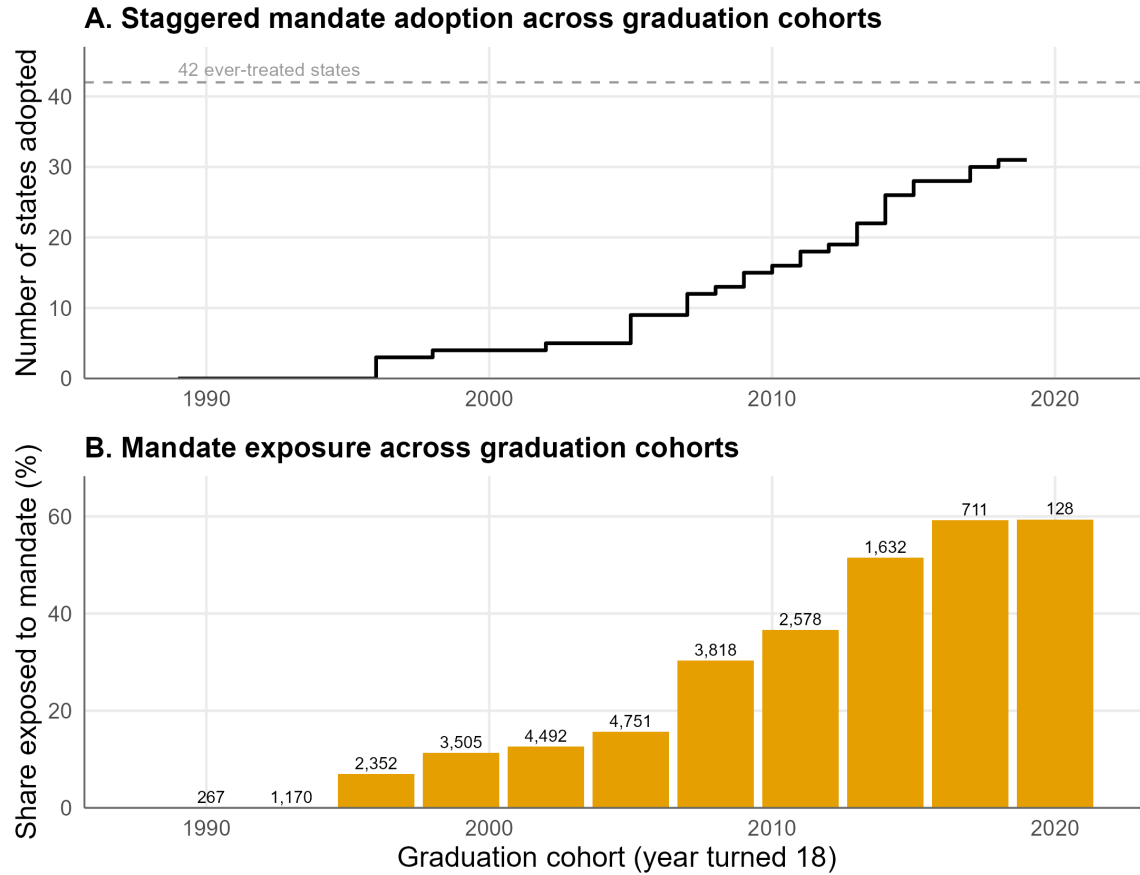
Table 8: TWFE estimates of the effect of high school personal finance mandates on financial knowledge

	Knowledge						Self-rated
	Score (0–5)	Comp. interest	Inflation	Bonds	Mortgage	Stocks	Score (1–7)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Mandate exposure $\times$ Female	0.13*** (0.04)	2.61* (1.49)	1.73 (1.59)	4.35*** (0.81)	0.13 (1.04)	4.28*** (1.30)	−0.03 (0.04)
Mandate exposure	−0.05 (0.04)	−1.32 (1.45)	−0.26 (1.40)	−2.74*** (0.72)	0.35 (1.04)	−1.14 (1.13)	0.02 (0.04)
1 if female	−0.40*** (0.03)	−5.93*** (0.62)	−11.67*** (1.05)	−7.08*** (0.45)	−3.26*** (0.70)	−12.11*** (0.73)	−0.43*** (0.02)
1 if non-White	−0.25*** (0.03)	−5.28*** (0.81)	−7.57*** (0.92)	−0.53 (0.48)	−8.37*** (0.86)	−3.39*** (1.24)	0.09*** (0.02)
Mean dep. var.	2.22	65.20	42.15	20.32	59.14	35.07	4.80
Observations	31,659	31,659	31,659	31,659	31,659	31,659	30,624
Adjusted R^2	0.076	0.032	0.051	0.011	0.042	0.040	0.049

Note: Estimates are from the same specification as Table 7. Column (1) is the number of the five Lusardi–Mitchell financial literacy questions a respondent answers correctly, from 0 to 5; columns (2)–(6) are those five questions taken one at a time, each an indicator equal to one for a correct answer. Column (7) is respondents’ assessment of their own financial knowledge, on a 1–7 scale. For the five binary items in columns (2)–(6), the mean dependent variable is reported as a percentage. Standard errors clustered at the state level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

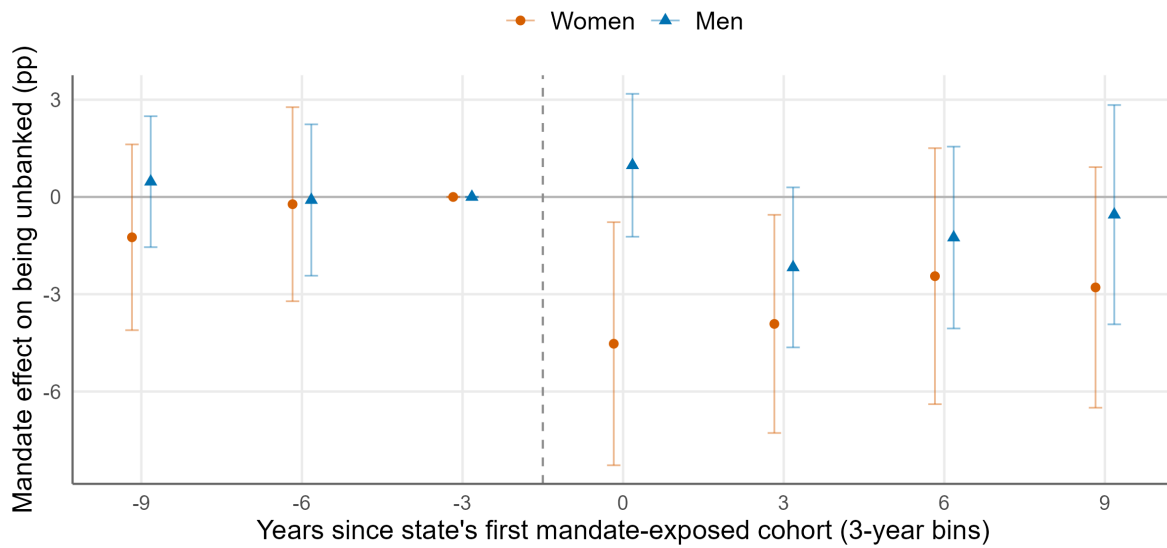
Figures

Figure 1: Staggered adoption of high school personal finance graduation requirements



Note: A graduation cohort is the year the respondent turned 18. Panel A shows the cumulative number of states whose requirement binds by each graduation cohort, and the dashed line marks the 42 ever-treated states. The count reaches 31 by the 2019 cohort because the remaining 11 states first bind on cohorts graduating in 2020 or later, outside the sample; they enter the estimation as not-yet-treated controls. Panel B shows the share of the analysis sample exposed to a mandate in each three-year graduation-cohort bin, with the number of observations above the bars.

Figure 2: Event study estimates of the effect of personal finance mandates on being unbanked by gender



Note: Estimates are from an additive extended two-way fixed-effects (ETWFE) specification with never-treated states as the comparison group, separately for women and men. Graduation cohorts are grouped into three-year bins, and estimates are normalized to the bin three years before the first exposed cohort. The horizontal axis is years since the state's first mandate-exposed cohort. Points are estimates in percentage points, and bars are 95 percent confidence intervals from state-clustered standard errors.

Appendix

Table A1: Sample means and estimated coefficients underlying the decomposition in Table 4

	Sample mean		Difference	Estimated	Contribution
	Women	Men	(Women – Men)	coefficient	to the gap
	(1)	(2)	(3)	(4)	(5)
Family income					+1.98 (0.15)
1 if family income is under \$15,000 (reference category)	0.217 (0.010)	0.131 (0.005)	+0.086 (0.008)		
1 if family income is \$15,000 to \$30,000	0.222 (0.007)	0.162 (0.008)	+0.060 (0.007)	−12.03 (1.06)	−0.72 (0.10)
1 if family income is \$30,000 to \$50,000	0.243 (0.006)	0.248 (0.008)	−0.004 (0.006)	−19.33 (0.93)	+0.08 (0.12)
1 if family income is \$50,000 to \$75,000	0.167 (0.005)	0.215 (0.006)	−0.048 (0.006)	−18.70 (0.95)	+0.89 (0.12)
1 if family income is \$75,000 or more	0.150 (0.014)	0.244 (0.018)	−0.094 (0.008)	−18.32 (0.86)	+1.73 (0.15)
Household structure					+1.75 (0.22)
1 if never married, no children (reference category)	0.505 (0.012)	0.811 (0.007)	−0.306 (0.010)		
1 if never married, with children	0.281 (0.009)	0.054 (0.005)	+0.227 (0.011)	+7.10 (0.78)	+1.61 (0.22)
1 if formerly married, with children	0.135 (0.007)	0.027 (0.002)	+0.108 (0.006)	+1.79 (0.82)	+0.19 (0.09)
1 if formerly married, no children	0.050 (0.003)	0.082 (0.005)	−0.032 (0.005)	+1.92 (0.85)	−0.06 (0.03)
1 if married (spouse absent), with children	0.018 (0.001)	0.003 (0.001)	+0.016 (0.001)	+3.43 (2.60)	+0.05 (0.04)
1 if married (spouse absent), no children	0.011 (0.001)	0.024 (0.002)	−0.013 (0.002)	+3.65 (1.46)	−0.05 (0.02)
Race and ethnicity					+1.06 (0.12)
1 if White (reference category)	0.479 (0.024)	0.566 (0.031)	−0.086 (0.012)		
1 if Black	0.266 (0.025)	0.160 (0.016)	+0.105 (0.012)	+10.08 (0.90)	+1.06 (0.13)
1 if Hispanic	0.163 (0.028)	0.160 (0.027)	+0.004 (0.006)	+4.11 (0.73)	+0.02 (0.03)
1 if other race or ethnicity	0.092 (0.011)	0.115 (0.014)	−0.023 (0.006)	+0.96 (0.71)	−0.02 (0.01)
Labor force participation					+0.42 (0.07)
1 if employed (reference category)	0.806 (0.006)	0.850 (0.007)	−0.045 (0.006)		
1 if unemployed	0.067 (0.003)	0.053 (0.003)	+0.014 (0.003)	+11.42 (1.26)	+0.16 (0.04)
1 if not in the labor force	0.128 (0.005)	0.097 (0.005)	+0.031 (0.005)	+8.59 (0.75)	+0.26 (0.05)
Age (in years)	29.019 (0.064)	28.767 (0.056)	+0.252 (0.050)	+0.22 (0.05)	+0.06 (0.01)
Foreign-born (1 if born outside the U.S.)	0.118 (0.014)	0.153 (0.015)	−0.035 (0.004)	−0.32 (0.64)	+0.01 (0.02)
Education					−0.23 (0.10)
1 if high school graduate	0.238 (0.009)	0.269 (0.012)	−0.031 (0.008)	+11.49 (0.81)	−0.36 (0.11)
1 if some college	0.353 (0.011)	0.309 (0.009)	+0.044 (0.008)	+2.98 (0.47)	+0.13 (0.03)
1 if college or more (reference category)	0.409 (0.016)	0.421 (0.015)	−0.012 (0.010)		

Note: I report the components underlying the decomposition in Table 4. Column (4) gives coefficients, in percentage points, from a linear probability model of unbanked status fit to the pooled sample with an indicator for gender (Fortin, Lemieux and Firpo, 2011). Each categorical coefficient is relative to the block's reference category, shown as its own row with columns (4) and (5) blank; age enters in years. Column (5) is each characteristic's contribution to the female–male gap, the product of columns (3) and (4) up to rounding, and within each block the contributions sum to the bold block total, which matches that block's row in Table 4. Refitting the pooled model without the gender indicator changes the explained share from 81 percent to 85 percent. The sample is 24,955 respondents (13,213 women and 11,742 men). Standard errors in parentheses are from 400 state-clustered bootstrap resamples.

Table A2: State personal finance mandates and first exposed graduating cohorts

State	First exposed cohort	Stand-alone course	In estimation sample
States with a personal finance graduation requirement (42)			
Illinois	1970	—	Yes
New Hampshire	1993	2027	Yes
New York	1996	—	Yes
Michigan	1998	2028	Yes
Wyoming	2002	—	Yes
Arizona	2005	—	Yes
Arkansas	2005	—	Yes
Louisiana	2005	2027	Yes
North Carolina	2005	2024	Yes
Georgia	2007	2028	Yes
Idaho	2007	—	Yes
Texas	2007	2030	Yes
Utah	2008	2008	Yes
Colorado	2009	2030	Yes
South Carolina	2009	2027	Yes
Missouri	2010	2010	Yes
Iowa	2011	2021	Yes
North Dakota	2011	—	Yes
Kansas	2012	2027	Yes
Indiana	2013	2028	Yes
Oregon	2013	2027	Yes
Tennessee	2013	2013	Yes
Nebraska	2014	2024	Yes
New Jersey	2014	—	Yes
Ohio	2014	2026	Yes
Oklahoma	2014	—	Yes
Minnesota	2015	2028	Yes
Virginia	2015	2015	Yes
Alabama	2017	2017	Yes
Maine	2017	—	Yes
Florida	2018	2027	Yes
California	2020	2031	No
West Virginia	2020	2028	No
Mississippi	2022	2022	No
Nevada	2023	—	No
Kentucky	2024	2030	No
Rhode Island	2024	2024	No
Connecticut	2027	2027	No
New Mexico	2027	—	No
Wisconsin	2028	2028	No
Delaware	2030	2030	No
Pennsylvania	2030	2030	No
States with no requirement (9)			
Alaska	—	—	—
District of Columbia	—	—	—
Hawaii	—	—	—
Maryland	—	—	—
Massachusetts	—	—	—
Montana	—	—	—
South Dakota	—	—	—
Vermont	—	—	—
Washington	—	—	—

Note: The first exposed cohort is the earliest high school graduating cohort required to complete personal finance instruction, whether in a stand-alone course or embedded in another required course. It typically follows the enabling legislation by several years, and it is the date the design uses to assign treatment. The stand-alone course column reports the first cohort required to take a dedicated course; a dash marks a state that meets the requirement through an embedded course instead. The last graduating cohort observed in the data is 2019, so a state enters the estimation sample only if its first exposed cohort is that cohort or earlier. Adopters whose requirement first binds afterward are marked No, and states with no requirement are listed separately, with a dash in every column. Adoption dates are from Urban et al. (2020) and Urban's compilation of state financial education requirements, last updated in 2024.

Table A3: Sample means of key variables by treatment status, FDIC survey sample

Variable	Total (1)	Ever-treated (2)	Never-treated (3)	Difference (4)
Banking status				
1 if unbanked	10.33 (0.54)	10.71 (0.57)	6.03 (0.91)	+4.69***
Household structure				
1 if never married, with children	17.21 (0.55)	17.38 (0.58)	15.32 (1.93)	+2.05
1 if never married, no children	65.02 (0.94)	64.56 (1.02)	70.21 (1.42)	-5.64***
1 if formerly married, with children	8.36 (0.45)	8.54 (0.49)	6.36 (0.54)	+2.18***
1 if formerly married, no children	6.61 (0.30)	6.72 (0.32)	5.37 (0.81)	+1.35
1 if married (spouse absent), with children	1.07 (0.06)	1.08 (0.06)	0.91 (0.14)	+0.17
1 if married (spouse absent), no children	1.72 (0.14)	1.71 (0.16)	1.84 (0.23)	-0.12
Race and ethnicity				
1 if White	52.05 (2.66)	51.82 (2.84)	54.61 (4.94)	-2.79
1 if Black	21.62 (2.21)	21.99 (2.34)	17.46 (7.32)	+4.53
1 if Hispanic	16.09 (2.74)	16.47 (2.94)	11.88 (1.66)	+4.58
1 if other race or ethnicity	10.24 (1.33)	9.72 (1.45)	16.05 (2.92)	-6.33**
Family income				
1 if family income is under \$15,000	17.61 (0.70)	17.90 (0.78)	14.38 (0.56)	+3.52***
1 if family income is \$15,000 to \$30,000	19.35 (0.75)	19.64 (0.81)	16.01 (1.40)	+3.64**
1 if family income is \$30,000 to \$50,000	24.55 (0.68)	24.76 (0.75)	22.11 (1.12)	+2.66**
1 if family income is \$50,000 to \$75,000	18.99 (0.42)	18.97 (0.46)	19.21 (0.83)	-0.24
1 if family income is \$75,000 or more	19.51 (1.63)	18.73 (1.80)	28.29 (1.94)	-9.57***
Education				
1 if high school graduate	25.49 (0.97)	25.86 (1.06)	21.26 (1.30)	+4.61***
1 if some college	33.15 (0.98)	33.51 (1.03)	29.09 (2.98)	+4.41
1 if college or more	41.37 (1.54)	40.63 (1.66)	49.65 (3.68)	-9.02**
Labor force participation				
1 if employed	82.59 (0.57)	82.39 (0.60)	84.81 (0.83)	-2.42**
1 if unemployed	6.03 (0.27)	6.13 (0.29)	4.91 (0.18)	+1.22***
1 if not in the labor force	11.38 (0.40)	11.48 (0.42)	10.28 (0.70)	+1.20
Foreign-born (1 if born outside the U.S.)	13.47 (1.52)	13.29 (1.67)	15.51 (1.58)	-2.22
Age in years	28.90 (0.05)	28.89 (0.06)	29.05 (0.14)	-0.16
Observations	25,404	21,214	4,190	

Note: I report survey-weighted means of respondent characteristics by treatment status. The Total column is the full analytic sample, and the Difference column is the ever-treated mean minus the never-treated mean. Standard errors clustered at the state level are in parentheses. Stars are from a weighted, state-clustered t -test of the difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A4: Sample means of key variables by treatment status, NFCS sample

Variable	Total (1)	Ever-treated (2)	Never-treated (3)	Difference (4)
Financial knowledge				
Quiz score (0–5)	2.22 (0.02)	2.19 (0.02)	2.38 (0.04)	−0.19***
1 if compound interest correct	65.20 (0.74)	64.14 (0.66)	70.79 (1.05)	−6.65***
1 if inflation correct	42.15 (0.71)	41.27 (0.70)	46.73 (1.19)	−5.46***
1 if bond prices correct	20.32 (0.39)	20.39 (0.44)	19.92 (0.73)	+0.47
1 if mortgage interest correct	59.14 (0.69)	58.65 (0.74)	61.70 (1.39)	−3.05**
1 if stock-vs-fund risk correct	35.07 (0.59)	34.32 (0.52)	39.01 (1.61)	−4.68***
Self-assessed knowledge				
Self-rated knowledge (1–7)	4.80 (0.02)	4.81 (0.02)	4.74 (0.05)	+0.07
Financial education				
1 if offered financial education (school, college, or workplace)	37.86 (0.44)	37.77 (0.46)	38.36 (1.26)	−0.60
1 if received financial education (school, college, or workplace)	24.08 (0.50)	24.06 (0.52)	24.15 (1.63)	−0.09
1 if received financial education in high school	14.85 (0.39)	15.06 (0.43)	13.79 (0.99)	+1.27
Demographics and education				
1 if non-White	55.06 (3.16)	54.47 (3.54)	58.18 (7.21)	−3.71
1 if aged 18–24	38.98 (0.63)	39.93 (0.45)	34.00 (2.01)	+5.93***
1 if aged 25–34	34.89 (0.48)	34.46 (0.44)	37.15 (1.72)	−2.69
1 if aged 35–44	26.13 (0.49)	25.61 (0.44)	28.85 (1.42)	−3.24**
1 if high school graduate	34.18 (0.83)	35.32 (0.72)	28.24 (2.27)	+7.08***
1 if some college	41.90 (0.79)	42.37 (0.65)	39.49 (3.55)	+2.87
1 if college or more	23.91 (1.30)	22.32 (0.88)	32.27 (5.65)	−9.95*
Observations	31,659	26,218	5,441	

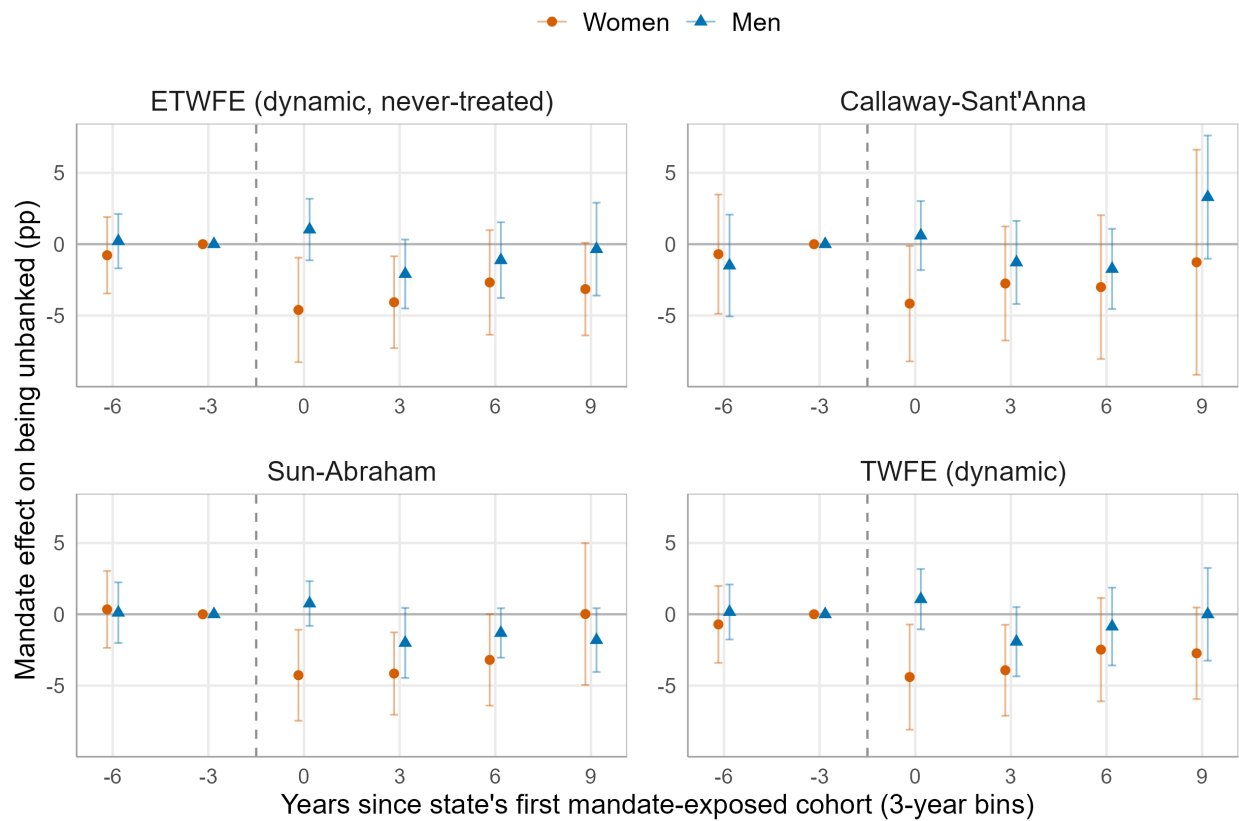
Note: I report survey-weighted means of NFCS respondent characteristics by treatment status. The Total column is the full analytic sample, and the Difference column is the ever-treated mean minus the never-treated mean. The financial education questions are available only in NFCS waves from 2012 onward. Standard errors clustered at the state level are in parentheses. Stars are from a weighted, state-clustered t -test of the difference. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: ETWFE estimates of the effect on the probability of being unbanked, adding post-treatment mediators as controls

	Dependent variable: 1 if unbanked				
	(1)	(2)	(3)	(4)	(5)
Mandate exposure $\times$ Female	-2.66*** (0.69)	-2.17*** (0.57)	-2.21*** (0.70)	-2.44*** (0.59)	-1.86*** (0.51)
Mandate exposure	-1.78** (0.76)	-1.53** (0.75)	-1.63** (0.66)	-1.89*** (0.70)	-1.64*** (0.62)
1 if female	4.71*** (0.52)	2.04*** (0.37)	5.35*** (0.51)	3.95*** (0.44)	3.03*** (0.36)
1 if Black	17.89*** (1.18)	14.54*** (0.97)	14.97*** (1.16)	15.98*** (1.12)	12.65*** (0.98)
1 if Hispanic	9.95*** (0.93)	8.62*** (0.85)	6.37*** (0.86)	9.47*** (0.84)	6.52*** (0.77)
1 if other race or ethnicity	3.12*** (1.20)	2.31** (0.92)	3.90*** (1.00)	1.87* (1.04)	2.26*** (0.81)
1 if foreign-born	-1.53** (0.66)	-1.43** (0.61)	-0.12 (0.58)	-1.44** (0.61)	-0.60 (0.56)
1 if family income \$15,000 to \$30,000	—	-15.77*** (1.11)	—	—	-12.44*** (0.96)
1 if family income \$30,000 to \$50,000	—	-24.09*** (1.13)	—	—	-18.47*** (0.92)
1 if family income \$50,000 to \$75,000	—	-25.25*** (1.13)	—	—	-18.00*** (0.88)
1 if family income \$75,000 or more	—	-25.65*** (1.13)	—	—	-17.18*** (0.87)
1 if Some college	—	—	7.97*** (0.50)	—	4.09*** (0.37)
1 if High school graduate	—	—	18.44*** (0.94)	—	12.82*** (0.74)
1 if unemployed	—	—	—	18.92*** (1.46)	11.25*** (1.08)
1 if not in the labor force	—	—	—	18.25*** (1.01)	8.79*** (0.79)
Mean dep. var. (%)	10.28	10.28	10.28	10.28	10.28
Observations	24,955	24,955	24,955	24,955	24,955
Adjusted R^2	0.066	0.162	0.124	0.119	0.201

Note: I include income, education, and employment here only to show how sensitive the estimates are to controlling for them. Because these variables are measured only after high school, the mandate may already have affected them. Conditioning on them absorbs part of the treatment effect and attenuates the estimated gender differential. For this reason, I study them as outcomes in Table 6. All columns are ETWFE estimates with adoption-cohort and graduation-cohort fixed effects, estimated on the subsample with all three observed, and coefficients are in percentage points. Standard errors clustered at the state level are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A1: Event-study estimates of the effect of personal finance mandates on being unbanked by gender across estimators



Note: Estimates are shown separately for women and men. Each panel gives a different estimator: dynamic extended two-way fixed effects (ETWFE), Callaway–Sant’Anna, Sun–Abraham, and dynamic two-way fixed effects (TWFE). The first three use never-treated states as the comparison group; dynamic TWFE uses the standard TWFE comparisons. The horizontal axis is years since the state’s first mandate-exposed cohort. Points are estimates in percentage points, and bars are 95 percent confidence intervals from state-clustered standard errors.